

A

Fragmented monitoring workflow

Heterogeneous observations

Displacement, settlement, pressure, water level, strain, and vibration

Project-bound model scripts

Implicit feature order, units, identifiers, and temporal assumptions

Disconnected alerts

Forecasts visualized without an audited formal-event boundary

Software gap

How can persisted forecasts become contextualized, inspectable, and conditionally executable records?

B

SHM-EM software boundary

1

Versioned engineering-semantic contract

Ordered features and targets
Units, transformations, artifacts, hashes

2

Synchronized Project Future State

One base time and future timeline
Observed and forecast risk remain distinct

3

Controlled forecast-to-event transition

Audited Evaluate; independently gated Execute
Formal event, response, and provenance

Backend-controlled formal side-effect boundary

Forecasting algorithm remains outside the contribution claim

C

Auditable engineering workflow

1 Configure

Objects, sources, rules

2 Forecast

Persist one versioned batch

3 Inspect

Completeness, series, Future State

4 Evaluate

Candidate plus audit run

5 Execute

Fresh Gate and rule recheck

6 Respond and trace

Formal records and evidence

Blocked eligibility produces no formal event side effects